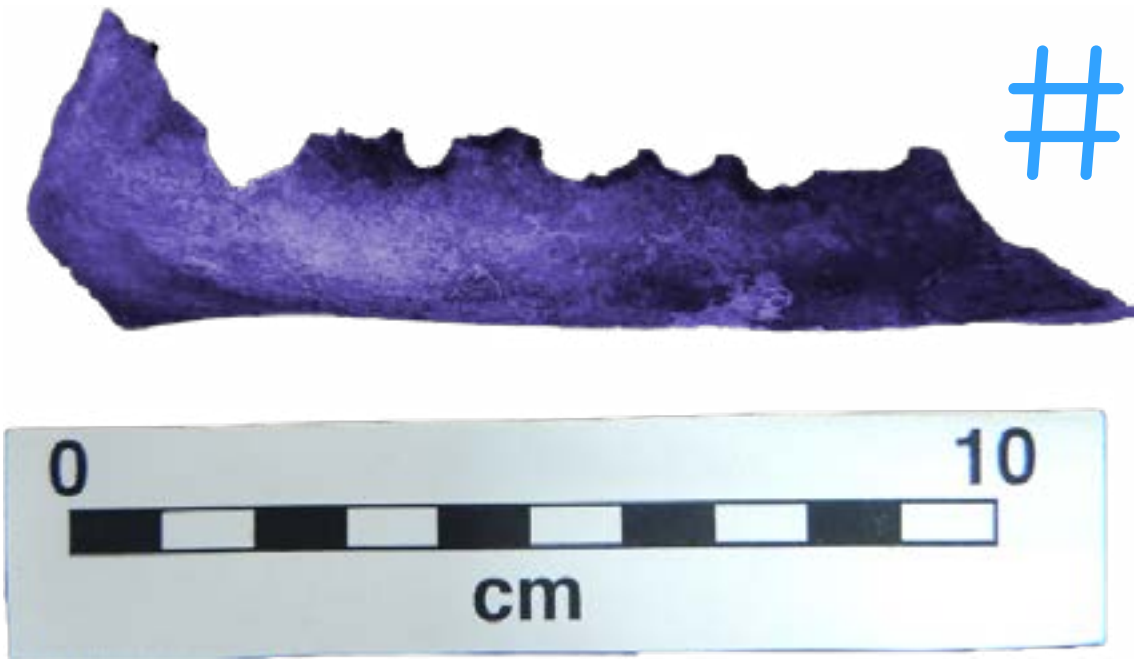

Guide Overview

1. Introduction
2. How to get started
3. A Moment in the Life of Jiă
4. The prompt list
5. Background on the *Oracle Bones in East Asia* data set
6. Sharing your work safely on social media
7. Acknowledgements
8. References

Estimated time to read this guide: 15 minutes

This Data Story contains no images of human remains.

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Introduction

Welcome (back) to the Digital Data Stories of the Alexandria Archive Institute and Open Context! This guide includes everything that you'll need to use archaeological data to inspire your next creative project. Whether this is for your own practice in archaeological illustration, to consider a theme for creating a fantasy setting set in the archaeological past for your gaming group, or to explore the material culture of a particular region for a class project, we can guide you through it. Of course, feel free to use whatever parts of this guide provide inspiration and get the creative juices flowing, or the gears turning, or whatever stuff we associate with starting to get things done, creatively.

This Data Story focuses on cultivating data literacy by creating art from data-inspired prompts using creative exploration. In this guide we include an introduction to the basics of this Data Story. We then include some examples of how to use the data-driven prompts and an example fictional narrative generated from our prompt list. This is followed by our creative prompts calendar. If you want to learn where these prompts came from, after the calendar, we provide a summary of the data set in the words of the people that published or produced the data set that inspired this prompt list. Finally, we explain how to safely share your work on social media then acknowledge the awesome folks who helped make this a reality and suggest further reading.

The guide can be used to celebrate or explore many inspirational projects. It can be used to work within the bounds of Artober or ArchInk which often coincide with National Arts & Humanities Month, Asian American Native Hawaiian Pacific Islander (AANHPI) heritage month, or even as inspiration during National Novel Writing Month (NaNoWriMo) or Archaeology Writing Month (ArchWriMo). In addition, feel free to integrate the prompts and the associated search tutorial into a class on any related subject or use it for personal enjoyment. All of the prompts in this list, and the guide itself, have been proofed, vetted, and sensitivity read, and so we hope that this provides a list to keep you inspired.

Accompanying this guide, we're also providing *Part Two: The Open Context Project Search Tutorial*. If the prompted word on our calendar isn't sparking joy or inspiration, you can use that term to search the project that generated the prompt list, the *Oracle Bones in East Asia* project, for images and other related information. This tutorial is specifically tailored to searching within that project.

If that's still not quite enough you can try searching Open Context (or the web) for more information and examples. If you need help searching Open Context outside of the original project, you can take a look at *The Open Context Search Tutorial* for how to conduct a keyword search on Open Context. With that, may the ancestors smile upon you and grant you inspiration.

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How to get started

As we mentioned, you can do anything you'd like with the prompts. But if that's too general, in this section we outline how to get started and an example of what we did with the prompts. If you don't need guidance and have a good idea of where you'd like to go, no problem, be inspired by data and embark on this journey.

Otherwise, look at the first word in the calendar and consider what it means to you. Consider if it's a noun or a verb and what experience you have had with it. Consider what you already know about the word. While we generated this list from an existing data set, you can think of your own experiences as data too. Your experiences with the word are the attributes or observations about it you can make. If nothing sparks your creativity just from thinking, you can use the *Part Two: The Open Context Project Search Tutorial*. This will bring you to the project that inspired this list.

There you can look for more information and photos to spark your interest. Searching around for more information is an important part of cultivating our reading and working with data literacies, so give yourself a good amount of time to do this. You'll want to read more about specific entries that came up when you searched the word, click through various photos, and possibly read some documents to provide more context. All of this will introduce you to new information.

If that doesn't work, or nothing comes up within the project search, try widening your search using *The Open Context Search Tutorial*. Hopefully between both of those, an idea will spark after looking and comparing the various data that relate to the prompt word. If not, consider searching the web more generally using the keyword. Once something interesting inspires you though, go with it! Your creations don't have to be huge, they can be small and simple too. This guide is here to support whatever your creative goals are and sometimes restricting a project helps us flourish!

You may wonder though, what happens if you get behind, or forget one day? While we organized the list to inspire a new creation each day, there's nothing stopping you from trying to use all the prompts at once or build on what you did on a previous day or combine the prompts from consecutive days. All of that is okay because we're interested in you developing your data literacy skills.

If you're not sure *what* your medium should be, one idea could be making soups or other foods inspired by the list, perhaps with a focus on spices associated with Chinese cooking. Or you could illustrate what it might mean to feel or be lucky. Or compose a short song about a trash pit. Or you can change what medium you do for each prompt. All of these are great ways to explore archaeological data and learn to work with it.

Below, we've provided an example narrative, based on the entire prompt list, to give you an idea of what can be done. Using all the prompts in one piece is also a possibility for solo practice or to utilize this guide within the bounds of a single educational session.

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A Moment in the Life of Jiǎ

Bones, bones, bones, that's all bàbā would talk about. The ***scapula*** we could prepare as tribute to be used in royal ***divination rituals***. Or that was the hope. If we were ***lucky***, the local lord would choose our bones, or maybe even our ***goats***, as tribute for the King, the ruler of the current ***dynasty***. Or that's what bàbā said anyways.

But before such pieces could be ***drilled*** properly and used in the ***sacred*** fires, they were just ***porous*** pieces. We had to use ***scrapers*** or other surfaces to remove the meat. Only then could they be ***inscribed*** with ***religious*** requests for the ancestors, or even just ***written*** on with our regional names, and have the fire applied to them to be part of this important process. Prepared bones were what a ***diviner*** of any caliber needed, not just a hunk of meat.

And in the ***present***, the failures of our ***household***, which didn't make it to the point of being good enough to be included as tribute, ended up in our ***trash pit***. Broken pieces, which had a ***crack*** or two before even getting heated, couldn't say anything about the future. Reminding us that the ***past*** is behind us and to not let our ***emotions*** get ahead of us.

I tapped the bone against my scraping surface and couldn't help but notice how ***hollow*** the ***sound*** was. Maybe this would be the one that the lord would pick. Perhaps even an ***auspicious*** sound though I tried to look at it with some ***objectivity***. I was new to the ***technology*** part, just having started apprenticing with bàbā a few months ago. But I felt confident that this was the one, even though it was the ***earliest*** in the process I'd noticed something unique about the piece, so I set it aside for now. The ***latest*** in my growing pile of bones that had gone through the proper ***treatment***.

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PART ONE: THE PROMPT LIST

 # Prompts are drawn from information in the Oracle Bones in East Asia data set.

1 BONE
2 DIVINATATION
3 PAST
4 ROYAL
5 WRITTEN
6 LUCKY
7 EMOTIONS
8 POROUS
9 INSCRIPTION
10 RELIGIOUS
11 CRACK
12 TECHNOLOGY
13 SCRAPER
14 OBJECTIVITY
15 AUSPICIOUS
16 SCAPULA
17 GOAT
18 SOUND
19 RITUAL
20 FUTURE
21 TREATMENT
22 HOUSEHOLD
23 DRILL
24 EARLIEST
25 FIRE
26 DIVINER
27 HOLLOW
28 PRESENT
29 TRASH PIT
30 LATEST
31 SACRED

RELEASE DATE: 07/2023
VERSION 1.0
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OPEN CONTEXT
THE ALEXANDRIA ARCHIVE INSTITUTE

Background on the *Oracle Bones in East Asia* data set

For this data-inspired creative project guide, we're drawing from a data set published on Open Context that is part of the *Oracle Bones in East Asia* project. The following is the summary for the data set on Open Context written by the people who contributed the data set. Further suggestions for general reading about oracle bones can be found at the end of this guide. Please note, though the contributers use the term 'East Asia' and the data set largely concerns finds within China, the research field of related scapulimancy extends across Asia. Within this Data Story, we have utilized similar vocabulary for consistency.

> Oracle bones — animal bones used for pyro-osteomantic divination rituals in East Asia — are one of the most important types of bone artifacts in Chinese Neolithic and Bronze Age sites and the source of inscriptions containing the earliest writing in ancient China. Although these inscriptions are the focus of most research, oracle bone use far pre-dates the inscribed examples and continues after they were a primary medium for writing. Uninscribed oracle bones are rarely published and there is a lack of metric data available for studying spatial and temporal trends in oracle bone manufacture and use. In the Oracle Bone Project, we are reviewing collections of oracle bones housed in institutions across China in order to collect comprehensive data on the types of animal bones used in divination, the methods of oracle bone manufacture, and the archaeological contexts in which the bones are found. Our goal is to trace the origins of oracle bone divination rituals, their spread across Asia during the Neolithic, and the ultimate development of oracle bone divination as a central part of Shang Dynasty royal religious practices. The project brings new zooarchaeological and technological perspectives to research on oracle bones and addresses Anthropological questions about the role of ritual technologies in household and state-level institutions.
>
> Data collected as part of the Oracle Bone Project is published on Open Context in a multi-language open access format. The raw data can be used by researchers around the globe to examine temporal and spatial trends in oracle bone manufacturing and use. Our focus is on uninscribed cases that have not received as much scholarly attention, but we encourage other scholars and institutions to upload additional data from inscribed or uninscribed oracle bones in their own collections. All contributions are associated with a publication record that is fully citable, searchable, downloadable in multiple formats, and linked to data standards that facilitate interoperability. Data input forms in English, Chinese, Korean, and Japanese are coming soon!
>
> The Oracle Bone Project is an international collaboration between the Institute of Archaeology, Chinese Academy of Social Sciences (IA CASS) and Harvard University.The project co-PIs are Katherine Brunson (Wesleyan University), Rowan Flad (Harvard University Department of Anthropology), and Zhipeng Li (Institute of Archaeology, Chinese Academy of Social Sciences).

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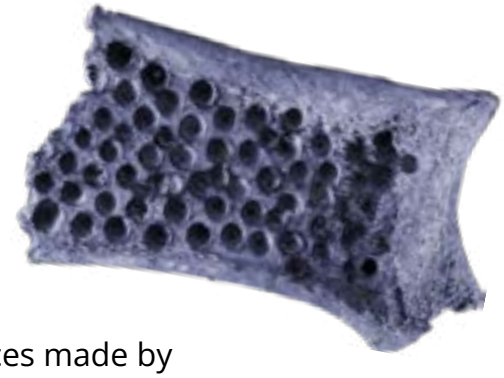
Sharing your work safely on social media

One way you might use this list is to participate in various online creative communities or thematic months via social media. These can be during thematic months inspired by particular heritages, such as AANHPI heritage month, or particular kinds of creative works, such as NaNoWriMo. There's even an archaeology themed tag on social media called ArchInk that runs in October. You can read more about that here.

And while the Data Literacy Project was in part inspired by ArchInk illustration, creative works in general are part of archaeological practice. In classrooms, students commonly learn to draw artifacts, make maps, or participate in ethnoarchaeological activities where we use our lived experiences to recreate the past. And those projects sometimes get shared online.

In addition, you might have just found this list and thought it was cool but might be thinking, what should I do with my creations? Well, once you've created whatever piece the day inspired, there are plenty of options. The first is always to just enjoy the piece for yourself and relish the fact that you created something based on archaeological data! You stretched your archaeological data literacy skills by taking a word and turning it into something new. Or you searched for more information about it and used that to play with the idea even more. And that is super cool all on its own.

Another option is to share your creation in some capacity. And that's where various hashtags and protections to your work might come in. Depending on whether you started this in association with a specific month there may be established hashtags that you'll want to use to associate your work with a specific community.



For example, the DLP (the "we" in this guide!) occasionally uses this guide to post the inspirational prompt daily during various months, encouraging interested participants to share their creations. We do this through sharing images (with alt text) and text. For Artober shares, these might include the prompt word as a hashtag, the day of the month, and the specific Artober or ArchInk tag you're participating in to associate it with other pieces made by folks focusing on archaeology-inspired creations. Exploring such tags is also a great way to see what other folks are creating and celebrate everyone's pieces.

If you choose to share your work, which is awesome and not mandatory, don't forget to include your signature, a watermark, or even a photo of yourself with your images. That way should anyone find and share it, you can get credited for your work. That way we can all enjoy the things we create while respecting people's rights to own their work.

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Acknowledgements

This Data Story is a collaboration with the producers of this data set, co-PIs Katherine Brunson (Wesleyan University), Rowan Flad (Harvard University), and Zhipeng Li (Chinese Academy of Social Sciences), and with our wonderful sensitivity reader Stefanie.

The writers of this data story, L. Meghan Dennis and Paulina F. Przystupa are not of East Asian descent. To ensure that we created a Data Story narrative that properly highlighted and respected the materials and the cultures associated with them, we reached out to some additional collaborators to ensure that this was an appropriate data set to use and that we did so in a way that respected the content highlighted in this data set. In addition, we'd like to thank Kate Ellenberger for thoughtful insights on the formalization of grassroots collaborations by existing institutional entities, which guided this Data Story into its current form.

References

For a quick but detailed overview of the archaeological (re)discovery of oracle bones and their purposes: https://en.wikipedia.org/wiki/Oracle_bone

For more information on scapulimancy: https://en.wikipedia.org/wiki/Scapulimancy

For additional images related to oracle bones, check out the MET: https://n2t.net/ark:/28722/k2959zk0r

For more more information on National Arts & Humanities Month, which runs in October: https://www.americansforthearts.org/events/national-arts-and-humanities-month

NATIONAL ENDOWMENT FOR THE HUMANITIES

Mellon Foundation

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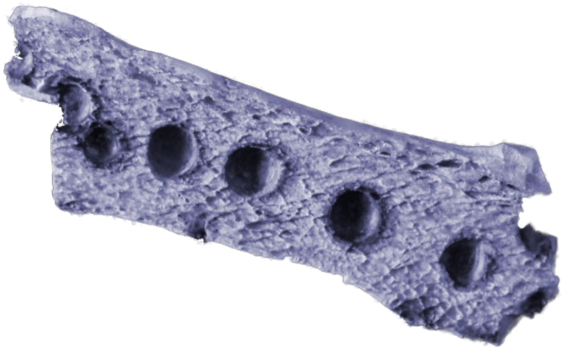
Tutorial overview

In Part Two of this Data Story we will learn to:

1. Navigate to Open Context on the web
2. Search or navigate to a particular project on Open Context
3. Look at data records from that project
4. Search those records using keywords
5. Examine digital objects associated with those keywords

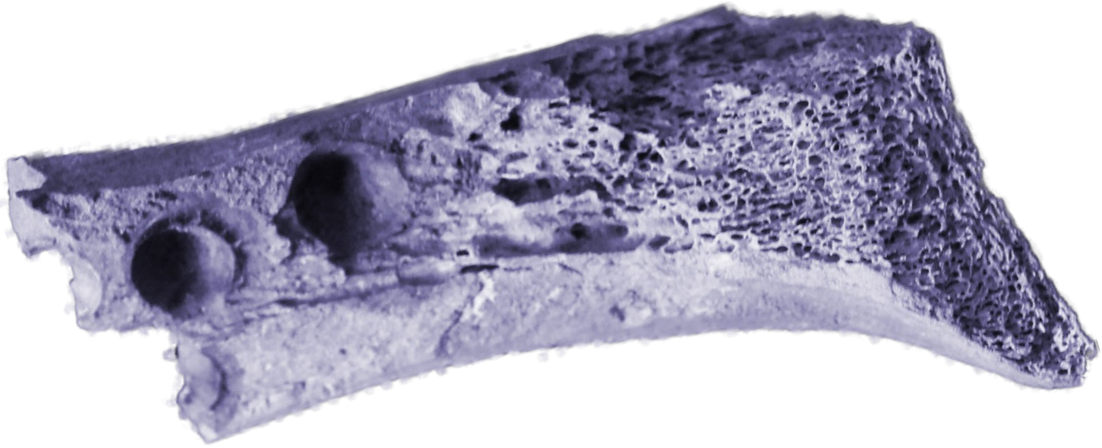
Estimated time to complete this tutorial: 30-60 minutes

Key concepts used in this tutorial



- browse for a project
- browse a data set
- data structures
- columns, attributes, headings
- entries, rows, observations
- objects
- keyword search

Estimated time to complete this tutorial: 60 minutes



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Introduction

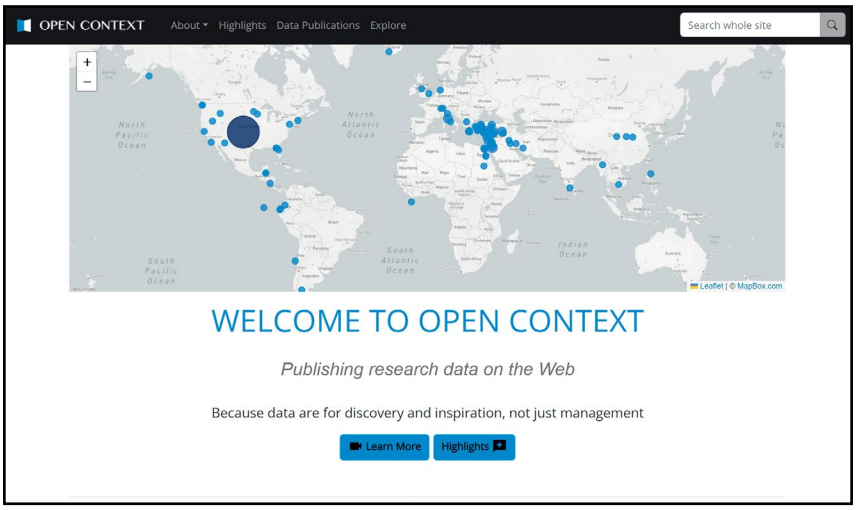
Welcome (back) to our Digital Data Stories! In this tutorial, we're going to focus on reading data. To do this we're going to walk through how to search for and access one specific data set on Open Context (OC). After that, we'll look at using images and textual information in that data set as artistic inspiration for something like Artober. If we're unfamiliar with Artober, check out *It's All in the Wrist (Bones): Archaeological Data as Artistic Inspiration Part One - The Prompt List*. But, why would we do this?

Well, we may be students assigned to use a specific data set on OC, or we may be interested in finding out about a specific project whose data OC holds. Or, maybe we thought the project sounded cool and we wanted to learn more about it as it seemed to fit with some ideas we're already working on. Regardless, we know what data set we want to explore, we just haven't had to find it on our own before. In this tutorial, we'll be navigating to and searching the data set called *Oracle Bones in East Asia*.

Finding Open Context on the Web

Likely, if we've found this data story we are probably familiar with searching on the web in some capacity. Or at least, we know which buttons to click to get to the sites we're familiar with. Regardless of how comfortable we are with getting to a website, we'll explore some options for how to navigate to Open Context (OC), where the data set we're interested in searching is located.

Once we've used one of the three options below for navigating to OC's website we can move on to the next section called **Finding our project**. Remember, our goal is to get to a page that looks like the one in the following image (though the color in the search bars might be different).



This is what the main page of Open Context usually looks like, and is where we'd like to go.

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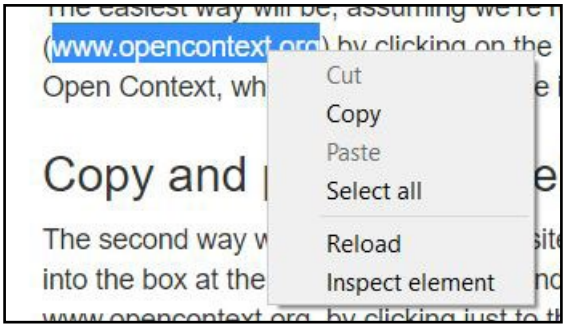
Clicking a hyperlink

The easiest way to get to the OC website will be, assuming we're reading this on a computer that is already connected to the internet, by going to https://opencontext.org. You can do this by by clicking on the words, "https://opencontext.org". This should bring up our default browser, and navigate directly to the main page of OC, which should look like the image we have above.

Copying and pasting a web address

The second way we can get to the website is by copying that website address or URL (which stands for Uniform Resource Locator) and pasting it into the box at the top of our browser window. To do this, we'll highlight the website listed in the **Clicking a hyperlink** section, https://opencontext.org, by clicking just to the left of the h and holding that click down while we use our mouse to scroll over to just past the **g** in "org". We will then use the opposite, typically the right, click button on our mouse to bring up a menu.



After opposite-clicking, these options will show up within a Microsoft OS environment.

We'll scroll down to "Copy" and click on that. We'll then open up our favorite browser. Then we'll navigate to the bar where the web address goes, typically an empty box near the top of the screen, and then click into that box to bring up the flashing cursor. We'll then opposite-click in that area bringing up that same menu. This time we'll scroll down to "Paste". That should put that URL into the box. If we didn't quite capture the URL correctly, we can also just type "https://opencontext.org" into that box.

Once the web address is in the box, we can usually hit enter, or click a variation of a "go" button and the browser will navigate to the OC main page as seen above.

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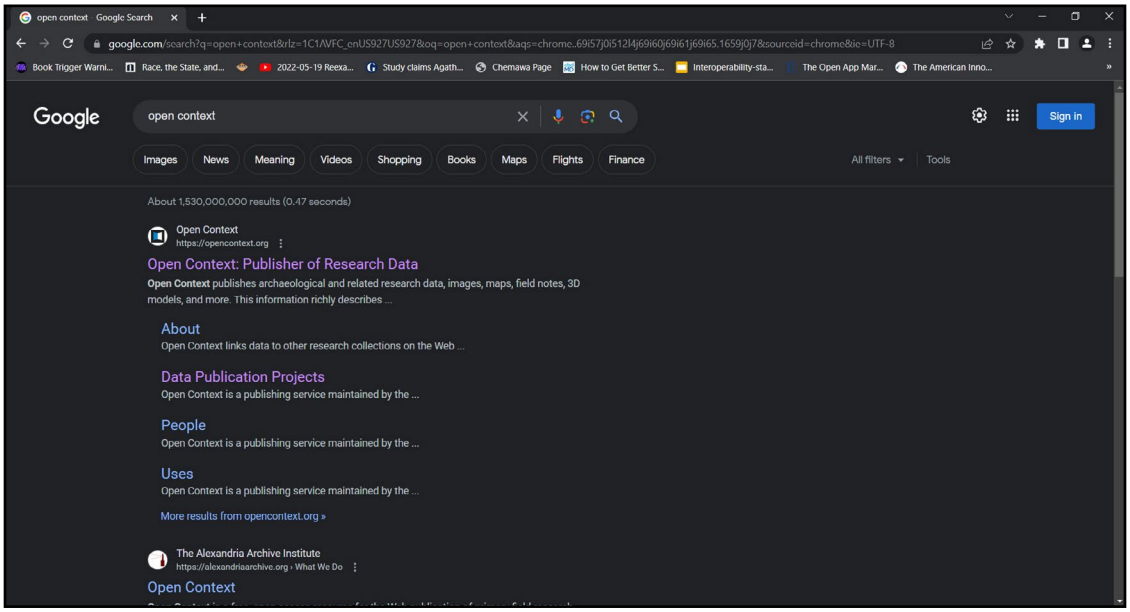
Searching for Open Context within our browser

One additional way we can navigate to the page is by directly searching for the OC website. We can do this by navigating to our preferred search engine or by typing "Open Context" into our web address box.

It's a lot more common now for browsers, the thing we use to interact with the internet, to have the box where we put in the website address or URL also double as a place to search, without having to navigate to a specific search engine. Which website will do the searching depends on the defaults within our preferred browser.

Typically, when we search we tend to toss the words or sentences we're interested in into the box. This is especially helpful if we don't remember exactly what we're looking for. So, if we didn't remember the whole website address for Open Context, we can just put "Open Context", on its own, into the box.

Once we have our favorite browser open, or have navigated to our preferred search engine, we'll just type "Open Context" into the search box and hit enter or "search" or whatever variant of that is appropriate for our online environment to navigate to the website.



After searching for "Open Context" we'll get a list that might look something like this (if we're using Google Chrome to search with Google though possibly in a different color).

Once we've hit that button, we'll probably have a long list of potential sites to go to. We'll be looking for the Open Context site that also has something about "Publisher of Research Data" associated with it. When we click on that link, we should get to a page that looks like the one at the beginning of this section.

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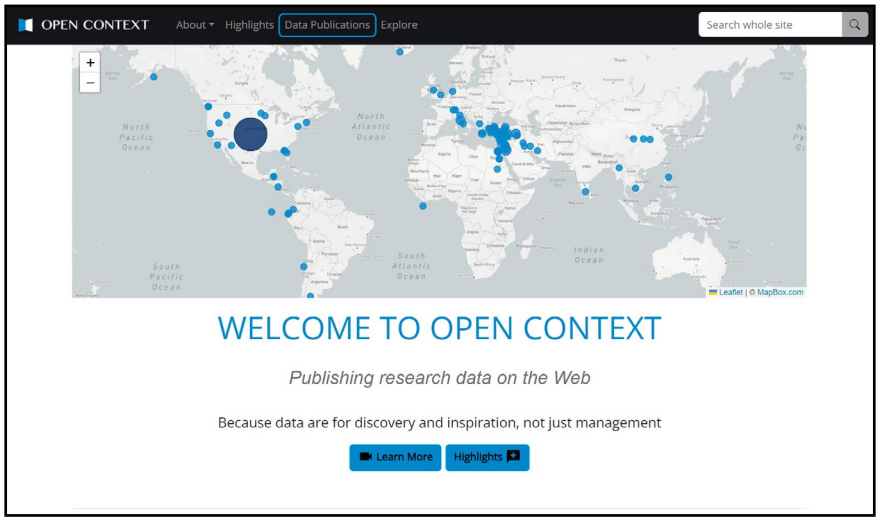
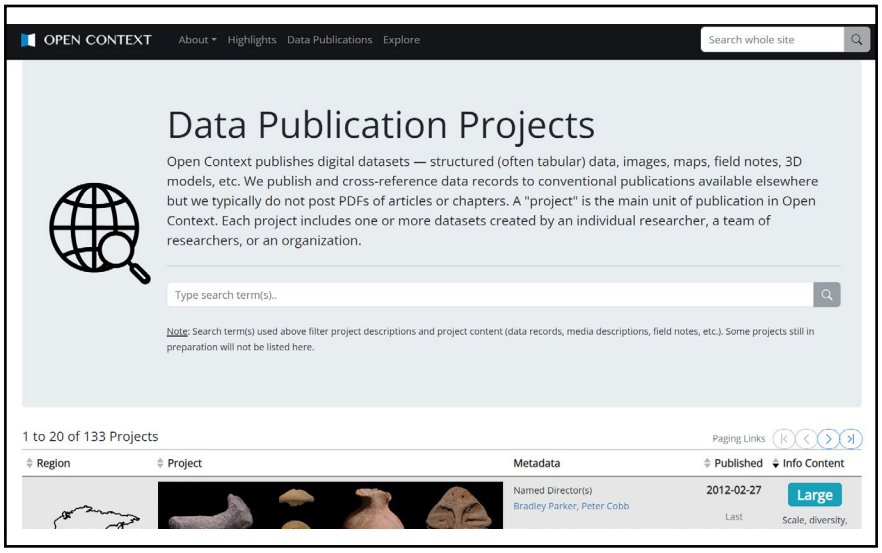
Finding our project

Now that we've located the Open Context website on the internet and navigated to it, we'll need to explore OC for the particular project we're interested in. We'll do this in a way that gets to the project directly. To start, we'll locate the "Data Publications" navigation option in the upper left portion of the main page, also known as the navigation bar and it should look like this:



This is where we can start looking for projects on the OC website.

Once we've clicked on "Data Publications" we'll be brought to a new page.



This is what the "Data Publications" page should look like after we click on it.

This is a collection of all the projects that OC currently publishes sorted by "Info Content, descending" as the default. We can change how projects are sorted by clicking on the little triangles next to each column heading.

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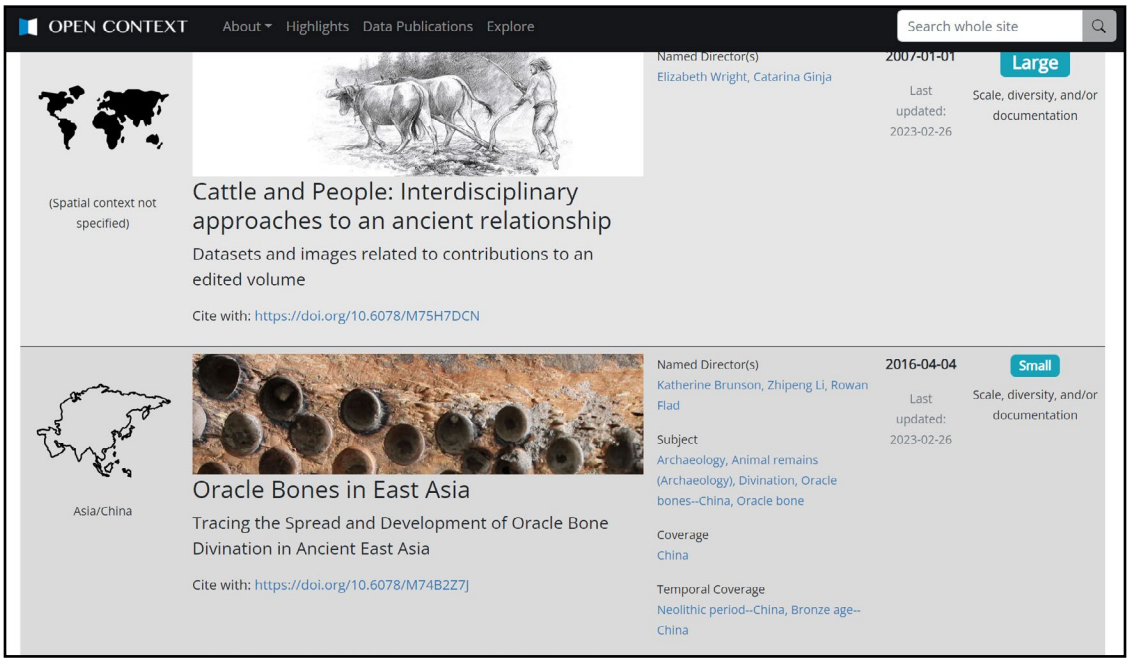
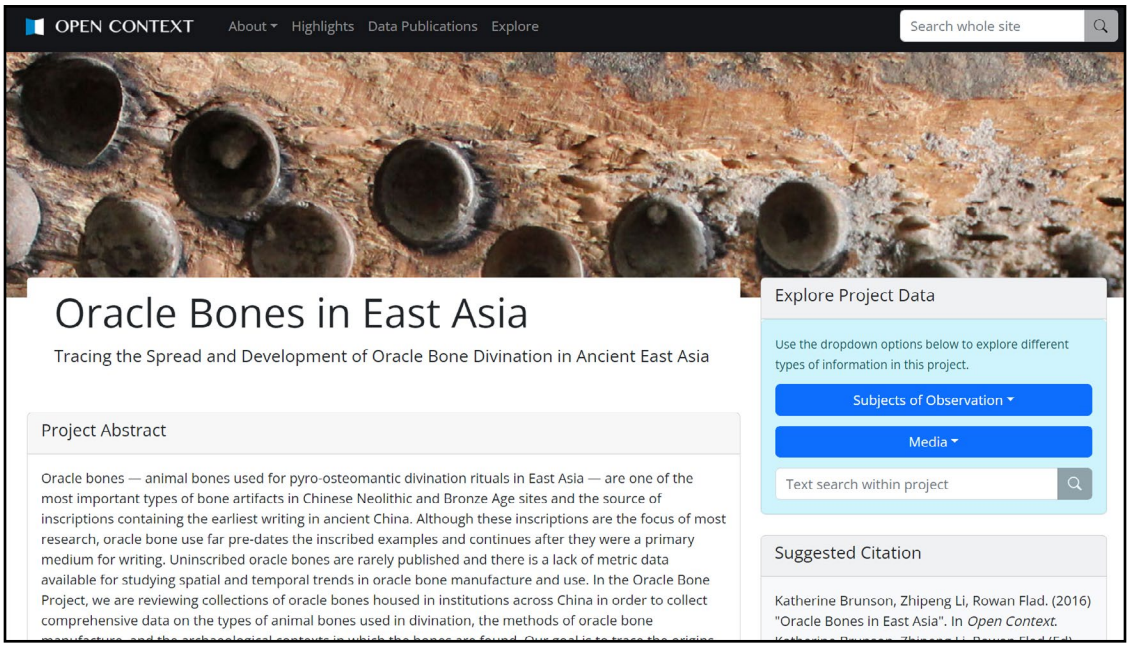
Since we already know that *Oracle Bones in East Asia* is a project published on OC, the fastest way to get to the project is to type in "Oracle Bones in East Asia" in the search box above the Projects table that says "Type search term(s)..". A slower way is to scroll to see if the project is in the first 50 entries and then keep going to the next page if it's not there.



This is the Oracle Bones entry we're looking for.

After finding the *Oracle Bones in East Asia* project, we'll click on the hyperlinked name of the project to bring us to the project specific page.



This is what the project page for Oracle Bones in East Asia looks like.

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This landing page gives us a description of the project overall and some background about who collected the data as well. If we're interested, we can spend some time reading this page to better understand the project.

Also, we've completed the first part of our search tutorial, we found the project! If we're feeling comfortable clicking around and seeing where this page takes us, great! Please feel free, that's what Open Context is for.

If we're using this guide in tandem with the *It's All in the Wrist (Bones): Archaeological Data as Artistic Inspiration Part One - The Prompt List*, we can take a look at the prompt for today and see where that takes us.

However, sometimes a little more guidance on how to search is important, and if that's the case, let's continue to the next section.

Searching and browsing the project data

Now that we've done all the hard work of locating the *Oracle Bones in East Asia* project, we can take a breather. Shake off any residual tension that comes from clicking, scrolling, and searching, and start playing around with the data from the project. We'd like to note that right now we can only access some of the data for this project, so if you search in the future you might find more data!

This means that as we're clicking through and exploring the data, we might not be able to open everything, and that's OK. Data take time to clean and get the right permissions to use, so it's important to have patience when using open data sets. To start exploring we'll begin by finding the "Explore Project Data" box on the Project page.

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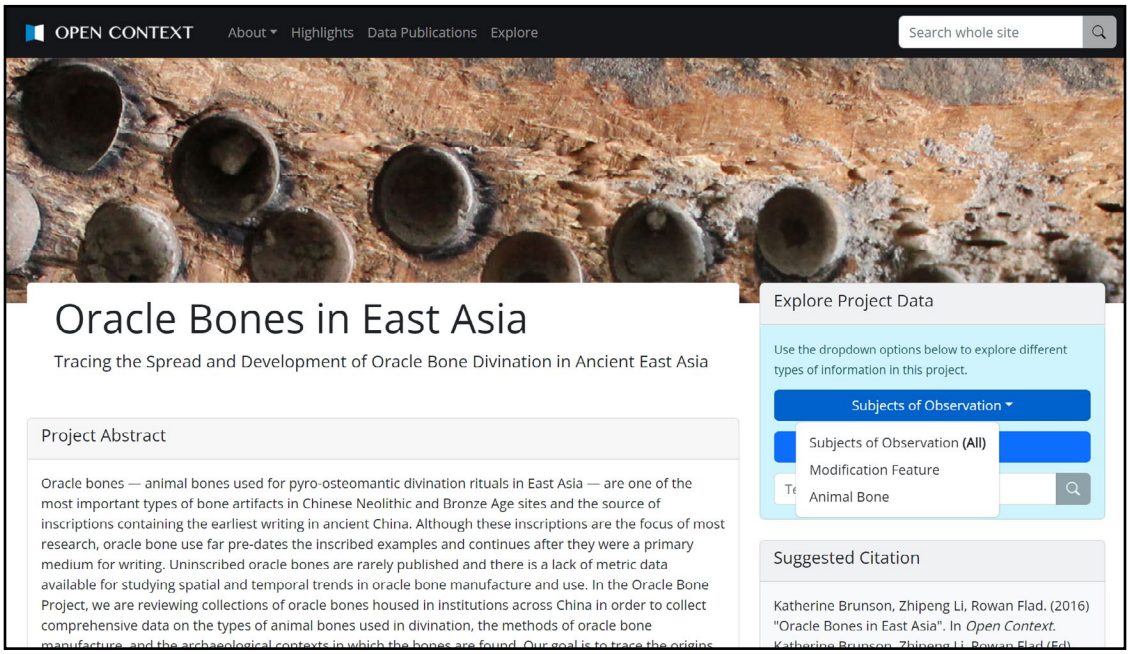
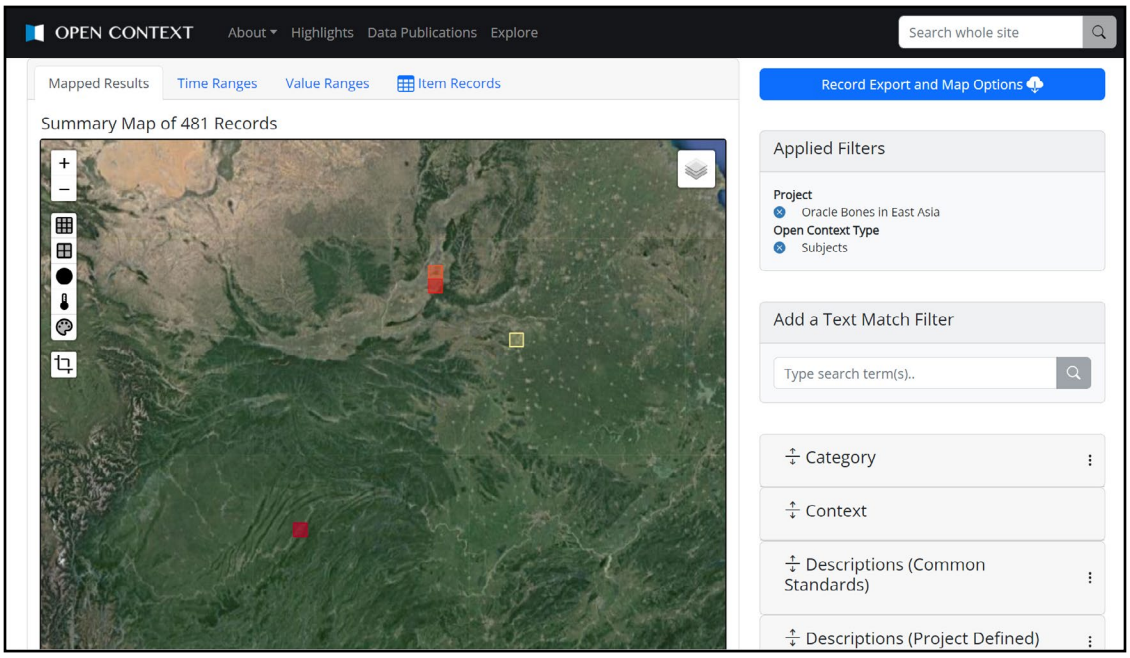
Browsing all data records



Once we click on the "Subjects of Observation" triangle we'll get a drop down menu that looks like this.

If we know we're looking for a particular kind of data we can click a specific option under the "Subjects of Observation" triangle menu. Since we're browsing for now, we'll want to scroll down to "Subjects of Observations **(All)**" and click there. This will bring us, once again, to a new page.



This is approximately the page we should have navigated to once we've clicked on "Subjects of Observations (All)" on the project page.

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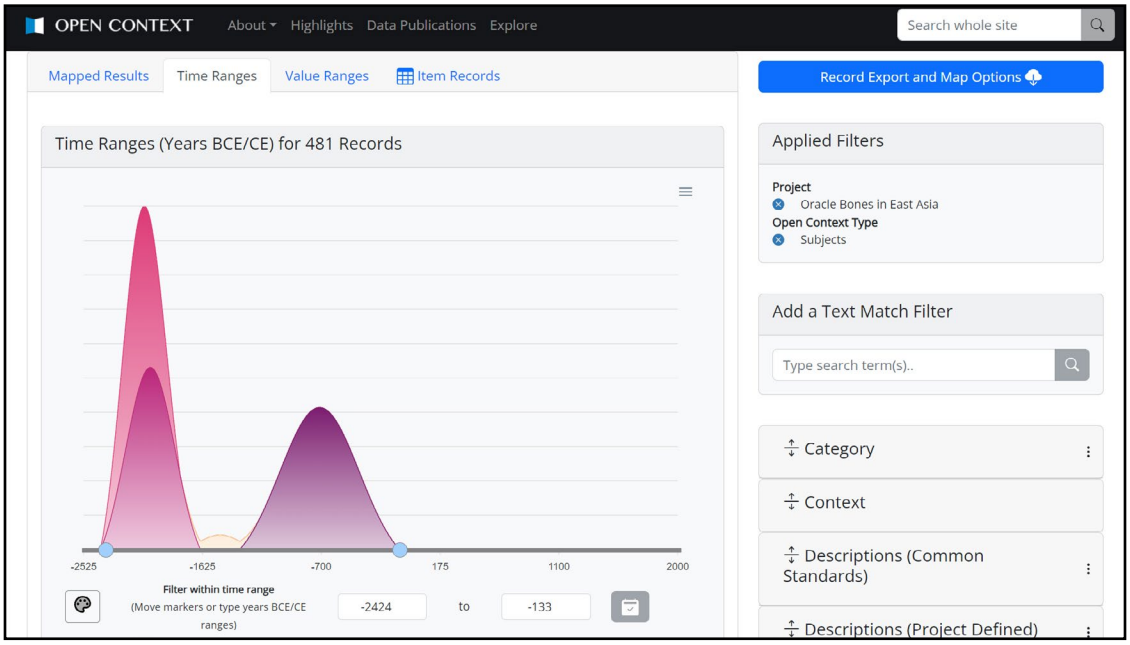
This brings us to a web page that lists all the records for this particular project. OC uses projects to organize the data it publishes. Each project represents the result of data creation work for research, conservation, and/or management by some entity. When data is systematically organized according to specific criteria (like a project), we call that data "structured".

Projects are just one way to structure data and each database, although similar in spirit, will have its own structures. This is important to remember if we explore other databases that might have their data organized into "galleries" or "collections". These are just different ways to define specific aggregations of what they have available and different choices for how to organize what they have available. However, the structure of data isn't the first thing we'll notice visually. Most likely, the first thing will be a map with different colored squares.

These show us the approximate location of where these data came from in geographic space and an approximation of how many records are associated with that area. If we'd like, we can just stay here and play around with the map. If not, we can look at another interesting graphics generated from the data by exploring the different tabs.



This is a temporal distribution of the belongings from this project, with higher peaks corresponding to more belongings coming from that period in time.

After clicking on the **Time Ranges** tab, we'll see an image that looks like the one above. It looks like a couple of superimposed hills, with each having a different height and slightly different color. High hills mean a lot of returned results associated with that time range and low hills mean that there aren't very many.

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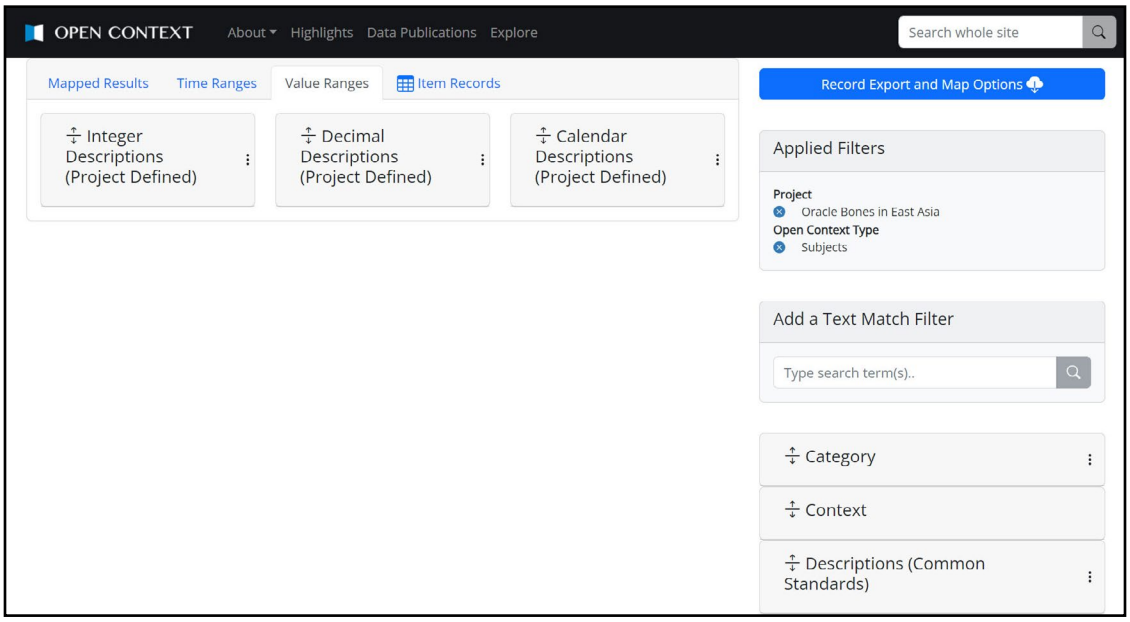
These hills also vary in width. Width indicates how sure we are of the date for a particular result and/or how long a period was. Wider hills indicate less surety, or larger range of potential dates, and narrower hills suggest better surety, or a smaller range of potential dates to associate with that result. The hills represent both how long something endured and an (educated) guess about the general time-frame associated with these results.

If we'd like, we can play around with some of the pieces on this graphic to see the range of times digital objects associated with the *Oracle Bones in East Asia* project appeared in the past. However, that can take a lot of time! So for those of us interested in other kinds of data, we can keep going through this tutorial and move to look at the the **Value Ranges** tab.



Value ranges are data driven options for simple visualizations of various measured attributes within the data set.

After we've explored the **Time Ranges** Tab, we'll move on to the **Value Ranges** tab. This tab suggests some attributes, pulled from the data, to visualize without downloading the data!What attributes options we have in this will change depending on the data present within a particular project or more refined group of data from multiple projects.

For *Oracle Bones in East Asia* the default Value Ranges that OC suggest are: Integer Descriptions, Decimal Descriptions and Calendar Descriptions. Under each categorization will be a list of specific attributes. We can either click directly on the attribute hyperlink or click the vertical ellipsis, select multiple attributes, and then the magnifying glass to visualize the data.

If we just clicked one attribute to visualize we'll get one bar chart (which we can learn more about by reading the *Cow-culating Your Data in Spreadsheets and R* Data Story). If wanted to look at multiple attributes we'll get multiple bar charts.

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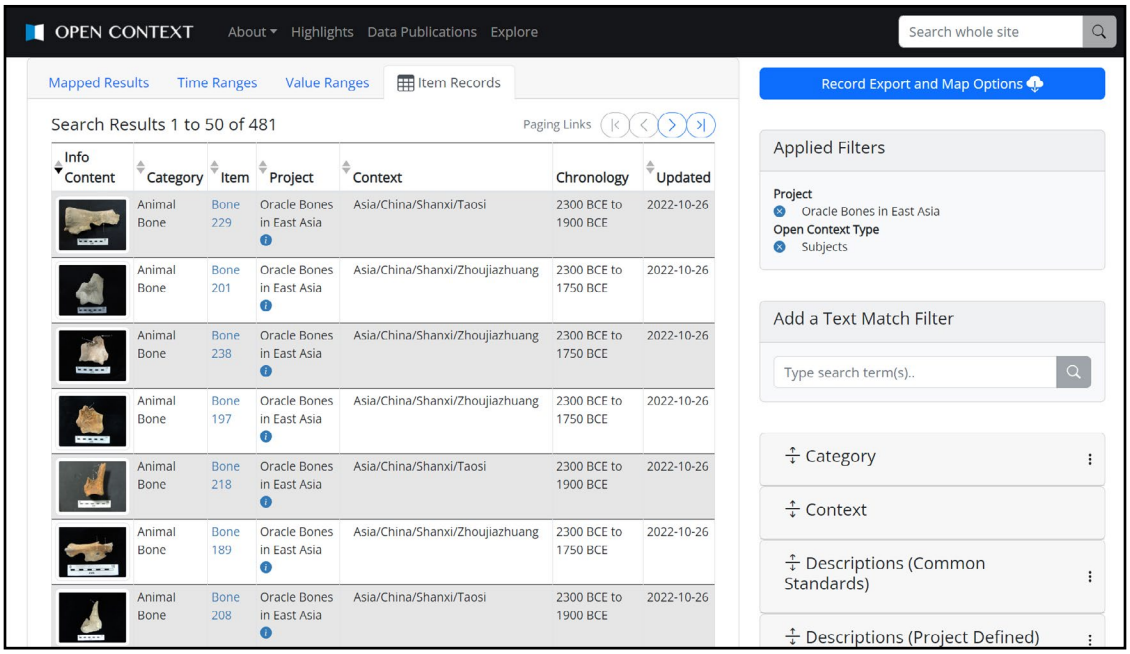
We can alter these bar charts by filtering the values using the "Filter Ranges" boxes on the bottom. We can do this by typing or clicking the up and down arrows to increase or decrease the values and then clicking the inverted triangle button to filter by these options.

While we've been playing around with these different projects and attributes we may notice that more filters got added to the Applied Filters box. These identify what specifically we're looking at and what records returned based on those filters. That's pretty handy to double-check exactly what we're looking at! But if we want to return to our original search, we'll need to click on the little circles with "X"s in them to get back to searching for information about 'bone'.



The last section of the "all data records" site includes a table of all the data records for this project.

After clicking through those other tabs, we'll click on the last one labeled "Item Records". It produces a structured arrangement of observations on the page. This is the actual list of data records for the project and it's organized into a table. If we aren't familiar with data tables we can take a look at the **Parts of a table** section. Otherwise, we can scroll through the existing list to examine what's there.

However, if we're feeling a little fatigued by all this scrolling and clicking we might choose to click on the first bone in the list. Clicking on your first entry will open a new tab rather than transform our current page.

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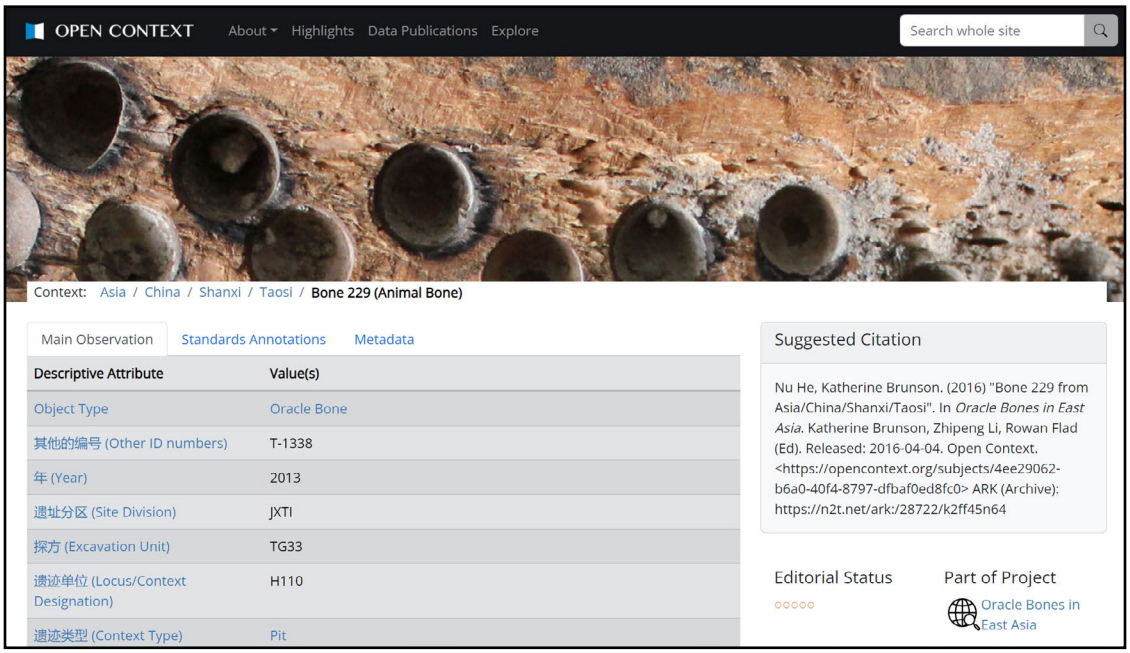

This is the unique page just for Bone 229 that we navigate to after clicking it's entry

Examining the record for this specific bone brings up all the information for it in it's own cool page. Looking at its photo and related information might be enough for inspiration, so we can stop here if we'd like.

But, if we'd rather get more detail about the objects, with the additional observations gathered for belongings from the *Oracle Bones in East Asia* data set, we can skip to the **Searching by keyword** section and start exploring the data using a specific criteria.

However, if we're still wondering how to read the data in the table we can take a look at the next section.

Parts of a table

So, tables aren't just things with four legs that support your laptop, tables are also a way to organize observations. In this kind of table, there are typically two axes of information that we use when structuring data into this form.

These are the horizontal headings, fields, columns, or attributes that list the specific observations that were taken for each object. For the list table on this page, these attributes are: Info Content, Category, Item, Project, Context, Chronology, and Updated. These are usually the first row, or line, in a data table.

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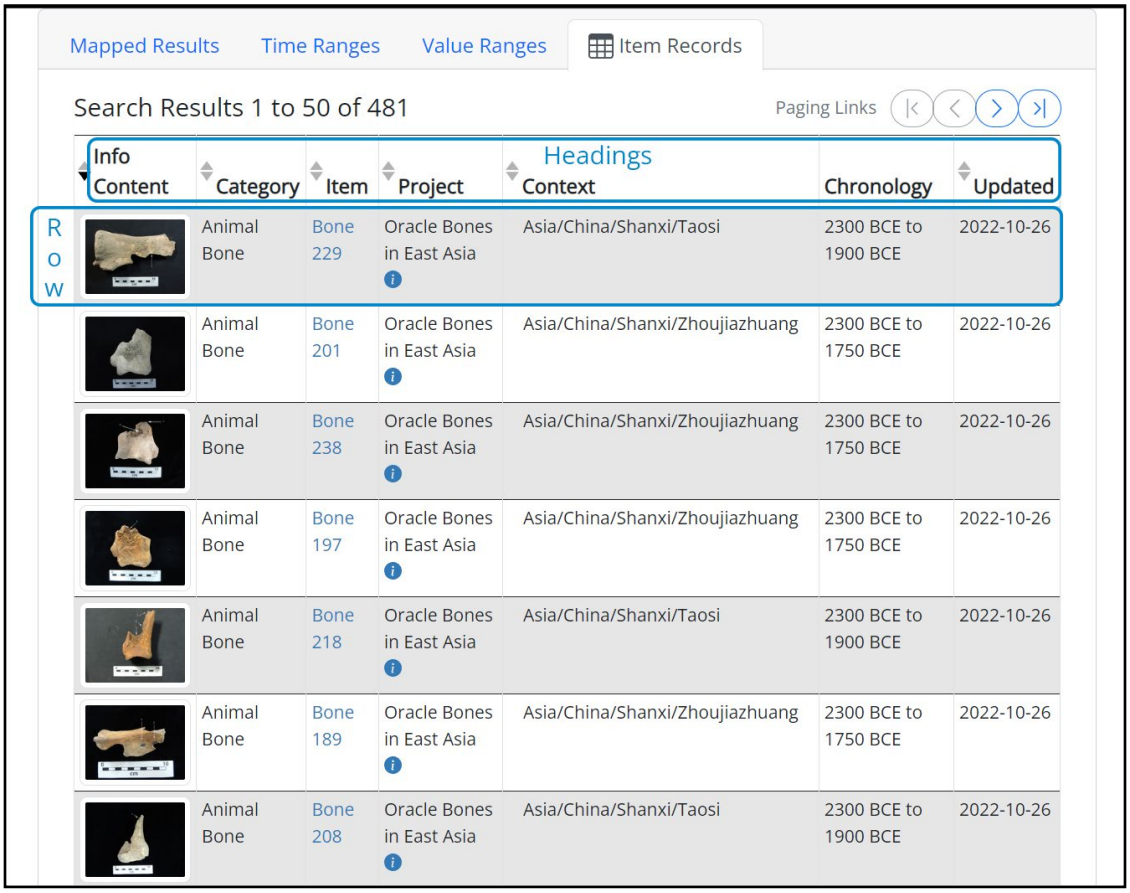

The data records in the "Item Records" tab are organized into a table. We've circled the headings and identified an individual row with a bounding box.

The next axis of information are the things listed vertically – entries, rows, or observations. These refer to the specific belonging we are observing, in this case a particular oracle bone, and the specificities of the attributes it had. These are listed so that everything in that line refers to the same object. When we searched this time, this page had the first entry as follows:

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Info Content:

Category: Animal Bone

Item: Bone 229

Project: Oracle Bones in East Asia

Context: Asia/China/Shanxi/Taosi

Chronology: 2300 BCE to 1900 BCE

Updated: 2022-10-26

It's important to note that the preceding digital object is just an example. That's because Open Context is always changing, adding new data and adjusting to accommodate dynamic search trends. That means the next time we search, we might get a different digital object as our first entry and that entry will have the same kinds of observations noted about it. Regardless, if we're inspired we can find out more about that digital object by clicking the entry! However, the observations in this table are really interesting on their own and we can probably use them to come up with a cool data-driven creation.

A short note on the word "object" (for those who are technically inclined)

The word "object" can mean a lot of things when we're talking about archaeological data. It can refer to a specific belonging that an archaeologist, descendant, or layperson found or analyzed during an archaeological investigation, but it can also be more than that.

In the realm of archaeological data, we can consider the records we produce, which list the things we observe about a belonging, as "objects". In this case, these might be things like site forms (explored in more detail in our *Gabbing about Gabii* Data Story), or things like photos, like the entry from Photo ID-054 in the next section.

Warning, very technical bit ahead! The hierarchy for the linked open data on Open Context relies on something called CIDOC-CRM. This is one (but not the only!) international standard for sharing cultural heritage information in an interoperable way. While we call these things objects, as sometimes we might pull up a PDF rather than a belonging when searching Open Context, the CIDOC-CRM would refer to these as a "Physical thing".

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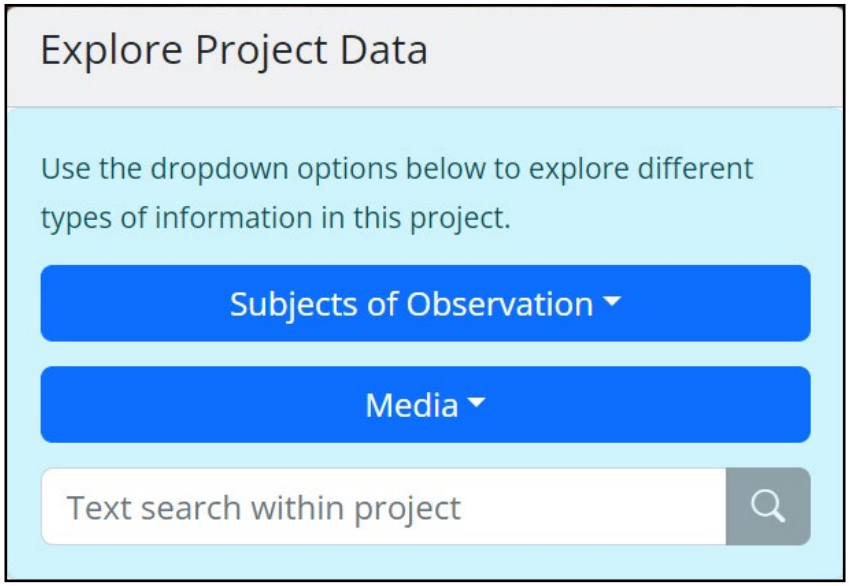
In Open Context, photos, PDFs, and data about individual belongings are "objects" because they refer to a single specific thing that has information about it stored in OC. In OC, each "thing" has a unique web address that acts as both a location and an identifier. This enables us to link to and cite specific objects, whether they're observations about belongings themselves or information about the observations made about belongings or sites.

In fancy terminology, these unique web addresses are called URIs. This stands for Universal Resource Identifier, giving a thing a name as well as a location. This is a little different than the URL, which is just the address or where a thing is located and is not the name of that thing.

Searching by keyword

If we know that tables are much more than places to put our plates or we've decided we just want to know about search, we'll start here. If after casually browsing all data we don't find what we're looking for, or if we know exactly what we'd like (perhaps because we just want to see what today's creative prompt brings up), we can approach the data in a different way. Instead, we'll do what is called a *keyword search*.

Similar to how we might have searched the internet for Open Context at the beginning of the tutorial, we can search within this project to find records related to a particular word. To do this though, we won't use the browser search bar, like to find a website. Instead, we'll use that box that we saw under the "Data Records" triangle in the "Browse, Search Project" box on the Project page.



The "Explore Project Data" box is found on the left side of the page.

If we've navigated away from the project page we'll need to click back to return to it or use this handy DOI, https://doi.org/10.6078/M74B2Z7J, to navigate directly there.

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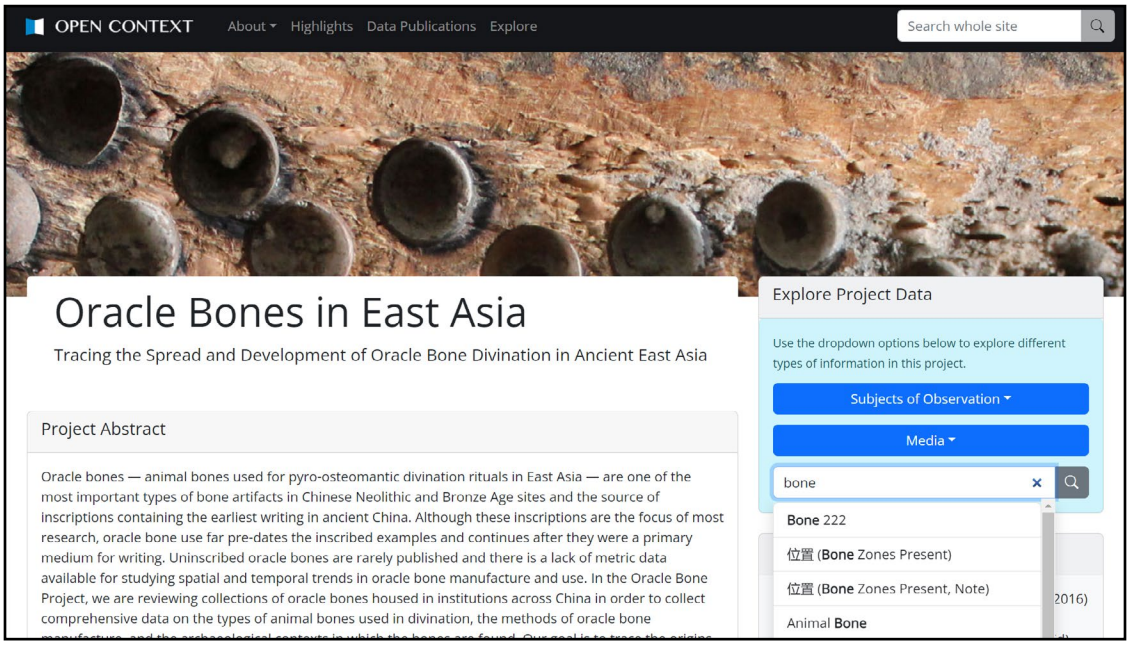
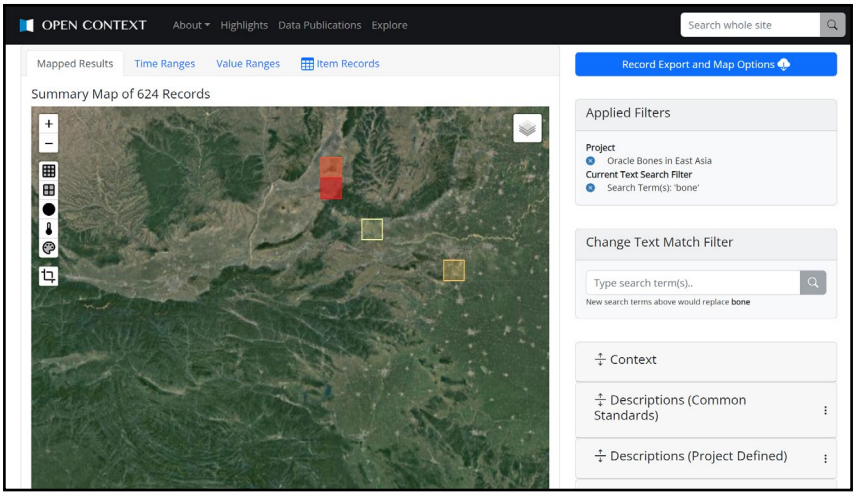
Once we've relocated the "Browse, Search Project" box on the Project page we'll find a bar within that box, below the options, that says "Search within Project". In that box we'll want to type a keyword. For this tutorial, we're going to go with the first word from the *It's All in the Wrist (Bones): Archaeological Data as Artistic Inspiration Part One* calendar and that word is "bone". We'll type that into the box, then click on the magnifying glass icon or hit the enter key on our keyboard.



Typing "bone" into the search box will look something like this.

If we completed the **Browsing all data records** section of this tutorial, the resulting page we're led to might seem very familiar. That's because it has the same structure as the "All Data Records" results page. However, rather than getting a map and a temporal distribution based on information from all of the data records, we get versions based on only the records that mention the word "bone" somewhere in their available information.



This is approximately the page we should get when we search for "bone" in the data.

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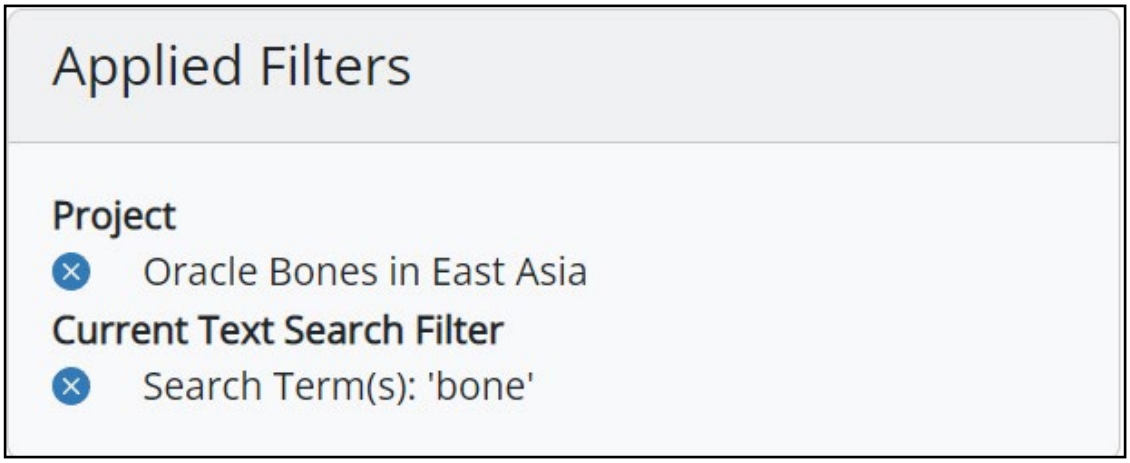
As these two pages look very similar, we might be worried we didn't do the right thing. However, we can easily check this. We can do this by finding a box called "Applied Filters".



The Applied Filters box should look like this.

Rather than just have the project name listed, this also has a filter called:

> **Current Text Search Filter**
>
> Search Term(s): 'bone'

This was the term we used to search. So we've definitely done the right thing! If we don't have this in that box, we'll want to go back to the tab for "View of Oracle Bones in Ea..." and type "bone" in again and see what comes up. Or we picked a different word to search so a different word is there, and that's totally cool.

By now, we're probably ready to see what information we got when we searched for "bone" in this project. To do that, we'll scroll past the map and the chronological distribution to the records list. If we aren't familiar with tables and skipped **Browsing all data records** we should go back to the **Parts of a Table** section within the **Browsing all records** section. If we're comfortable with tables we can start looking at what our search provided us with.

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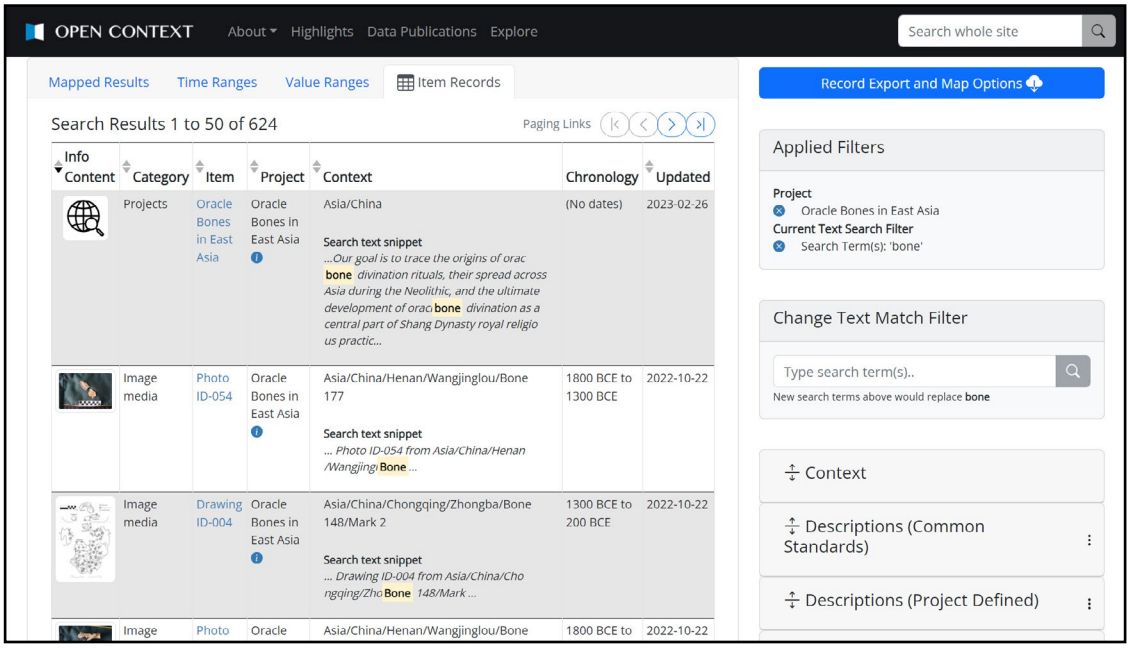

This is a slightly different list than the "all records list" as it only includes entries with "bone" in the record.

We'll see that in comparison to the table created when we looked at all records, this table has the same headings but a different object is listed first. Depending on when we search to try out this tutorial, our first entry might also be different so we'll just search for the same kinds of things in our results. In this table, the first non-project entry we get is for "Item Label Photo ID-054" and under "Category" we might notice that the word "Bone" is highlighted in the phrase,

> "China/Henan/Wangjinglou/Bone 177 ...Photo ID-054 from China/Henan/Wangjinglou/ Bone 177 Image Type Photograph Neolithic period..."

This identifies where the word "bone" was found in relation to this entry and specifically that it had to do with the category. Additionally, if we click the hyperlink for this record we go to a photo and page that has more information about that object. And that's pretty cool.

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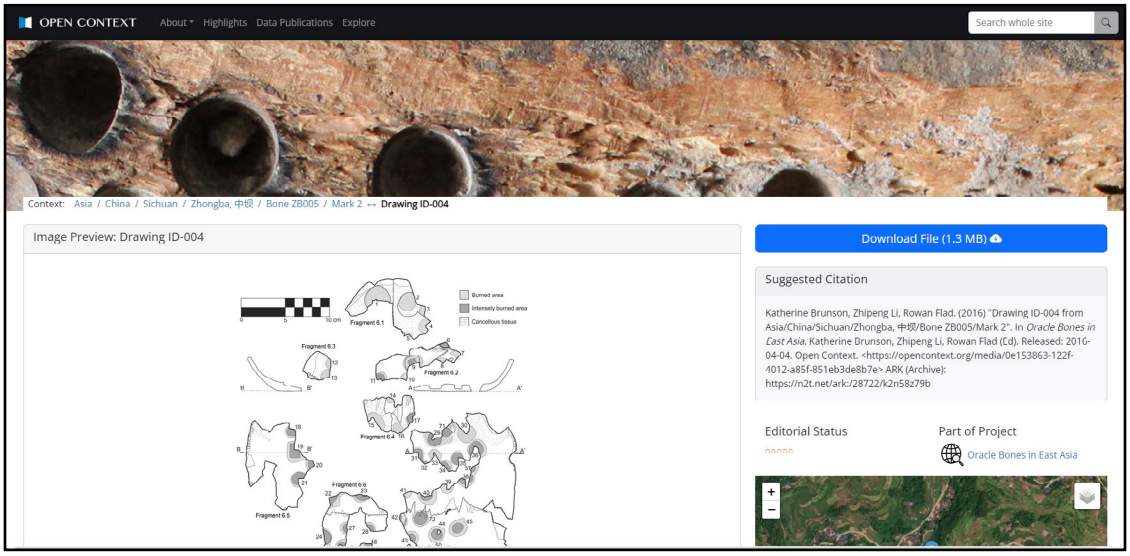

This is the page for Drawing ID-004 and near the top we'll notice a <-> symbol that associates this photo with Mark 2.

Whether we click on the hyperlink for the name of the item or on the thumbnail photo associated with that entry, we'll be brought to the same page. This page is for the specific photo and if we look near the top of the page, just above the enlarged image, we'll see a <-> image, an arrow with pointers on both ends, that suggests that this image, besides being its own unique object, is associated with Mark 2.

This is a close up of the double arrow <-> symbol that we are looking for. It indicates a relationship between the two pieces.

Awesome! This means that there's more information about this photo (and the bone in it) that we can explore. If we'd like, we can click on that name and read more information about Mark 2 itself rather than just information about its portrait.

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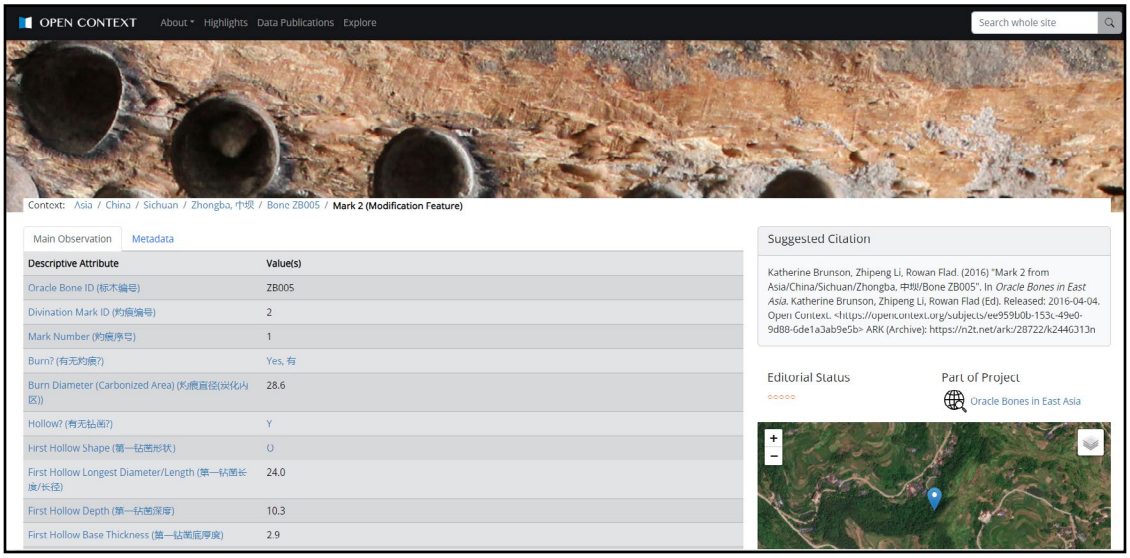

The page for Mark 2 should look like this.

Once we navigate to the page for Mark 2 we can see that there is a lot more information about that digital object to use as inspiration. This includes other photos associated with it as well as specific descriptions of its attributes. Congratulations! We've found one belonging (digital object? artifact?) that we can use as inspiration for today's search prompt, or that can teach us more about oracle bones.

When we get a new prompt for tomorrow, or if we just want to look at more oracle bones, we can use the same process to search for more belongings around that theme or around any other keyword of interest!

Conclusion

Congratulations! We learned to navigate to the Open Context webpage, explore projects, find a specific project by name, navigate to the project page, and then browse the data in a couple of different ways. That's a lot to learn, but there is also a lot of cool stuff out there about the oracle bones just within the project's information.

Now that we know how to do that, we can use this same process to search for the other prompts in our creative list, if we're participating or utilizing the *It's All in the Wrist (Bones): Archaeological Data as Artistic Inspiration Part One - The Prompt List*, and see what sparks our imagination. We can keep browsing or searching and do so for as long as we'd like. Additionally, it's important to practice these skills so that navigating in this fashion becomes more comfortable. Therefore, even if we think we've got this down, we may want to come back again, particularly with a different search terms, to see what happens.

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What will we do if there are no records? Where will we find inspiration? Well, we might need to try something else if that's the case, but let's be happy we've found what we have so far. With that, we can enjoy exploring the rest of the prompts in our creative list, if that's why we're here, and find out more about oracle bones in general by checking out the *It's All in the Wrist (Bones): Archaeological Data as Artistic Inspiration Part One - The Prompt List* and generally enjoy what the archaeological data records on OC inspire in our own work.

Key concepts used in this tutorial

Browse for a project

Scrolling through a list of projects to find one we're interested in.

Browse a data set

Looking through a particular data set to find something that's of interest to use.

Data structures

Organizing data according to the desires of the institution hosting that data.

Columns, attributes, headings

Specific observations that were taken for a particular object.

Entries, rows, observations

Specific things that were observed, and the particular instance of the attributes they had.

Objects

Defined entities within a particular data structure (belongings, documents, and images).

Keyword search

Using a particular word to search the web or a project within OC or any other searchable place, looking only for entries that include that particular word.

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